Self-Referential Spectral Action: Existence and Nontriviality of Fixed Points

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Abstract

We investigate whether the Dirac operator in Connes' spectral action framework can depend on its own spectral action value, forming a self-referential loop $D = D_0 + \Phi(S[D])$. For the scaling ansatz $\Phi(S) = g(S) \cdot D_0$ with $g(S) = \kappa \ln(S/S_0) + \beta$, we prove that the composite map $F : S \mapsto S[D_0 + \Phi(S)]$ is a Banach contraction on a neighborhood of the unperturbed action S_0, guaranteeing a unique fixed point S^* whenever $|\kappa| < 1/10$ and $\beta \neq 0$. The fixed point is nontrivial ($S^* \neq S_0$) for generic β.

The central physical result is the following: the self-referential structure determines a self-consistent conformal factor $\lambda^* = 1 + g(S^*)$, under which all standard model quantities scale by powers of λ^*. The dimensionless ratio

$$\mathcal{R} = \frac{\Lambda_{\mathrm{cc}} \cdot G_N^2}{g_{\mathrm{YM}}^4 \cdot v^4}$$

is **invariant under λ^*-scaling**, and therefore independent of the free parameters κ and β. This $\mathcal{R}$-invariance is the unique robust prediction of the self-referential framework that survives the "scalar bottleneck" (one equation $\to$ one constraint). All other results—including the existence and nontriviality theorems—depend on the choice of Φ and its parameters, which currently lack a physical derivation.

Keywords: spectral action, noncommutative geometry, Banach fixed point, self-referential, conformal invariance

1 Introduction

1.1 Motivation

In standard physics, the action functional $S[\varphi] = \int \mathcal{L}(\varphi, \partial\varphi)\, d^n x$ is externally given—the Lagrangian and its parameters come from "outside." In a companion work [16], the first author introduced the concept of a *self-referential action* satisfying $S[\varphi] = F_\varphi(S[\varphi])$, where the action value appears as an argument of its own functional. Using the Banach fixed-point theorem on a compact manifold, existence and nontriviality (in the sense of inequivalence to any standard action under jet-local diffeomorphisms) were established.

The present work investigates whether a similar self-referential structure can be realized within Connes' noncommutative geometry (NCG) framework, specifically at the level of the *spectral action*.

1.2 The Spectral Action Principle

Connes' spectral triple $(\mathcal{A}, \mathcal{H}, D)$ provides a geometric framework that reconstructs the Standard Model coupled to general relativity from purely spectral data [1, 3]. The spectral action

$$S[D] = \mathrm{Tr}\ f(D/\Lambda) + \langle \psi, D\psi \rangle$$

where f is a positive cutoff function and Λ is the cutoff scale, encodes the Einstein-Hilbert action, Yang-Mills kinetic terms, the Higgs potential, and fermion couplings in a single trace [2].

However, in the standard framework, the Dirac operator D (and hence the 25+ Standard Model parameters it encodes) is *externally given*. The spectral action evaluates $S[D]$ for a fixed D; it does not constrain D itself.

1.3 The Core Question

Can D depend on its own spectral action value? Formally, we ask whether there exists a map $\Phi : \mathbb{R} \to \mathrm{End}(\mathcal{H})$ such that

$$D = D_0 + \Phi(S[D])$$

and the composite map $S \mapsto S[D_0 + \Phi(S)]$ has a fixed point. If Φ is a contraction, the Banach fixed-point theorem guarantees a unique self-consistent S^*.

1.4 What This Paper Predicts—and What It Does Not

Hard results (independent of κ, β values):

- Existence and uniqueness of the fixed point S^* (Theorem 3.8)
- Nontriviality $S^* \neq S_0$ (Theorem 3.9, conditional on $\beta \neq 0$)
- **Invariance of the dimensionless ratio $\mathcal{R}$** (Proposition 4.1)—the core physical prediction

Soft results (depend on κ, β, which are currently free parameters):

- The correction magnitude $S^*/S_0 - 1 \sim -4\beta/(1 + 4\kappa)$
- The specific value of λ^*
- Absolute corrections to individual physical quantities

This paper cannot:

- Predict the absolute values of the 25 Standard Model parameters (scalar bottleneck)
- Determine κ and β from physical principles
- Resolve the cosmological constant problem (correction is $O(\beta) \sim O(10^{-2})$, far short of 10^{183})

1.5 Relation to Paper I

1.6 Outline

Section 2 reviews the mathematical framework and **verifies the heat kernel coefficients against CCM06**. Section 3 defines the self-referential spectral action and proves the main results. Section 4 discusses physical implications, **with $\mathcal{R}$-invariance as the centerpiece**. Section 5 addresses limitations, **including an honest assessment of Scheme B's physical motivation**. Section 6 reports the **INSPIRE-verified literature gap**.

	Paper I	This work
Self-reference	$S[\varphi] = F_\varphi(S[\varphi])$	$S[D] = S[D_0 + \Phi(S[D])]$
Variable	Functional $S[\varphi]$	Scalar $S \in \mathbb{R}$
Fixed-point space	$\mathbb{R}$ (for each φ)	$\mathbb{R}$
Nontriviality	Theorem (jet-local inequivalence)	Conditional (depends on Φ)
Core prediction	Rich (functional relation)	Single ratio $\mathcal{R}$

2 Mathematical Framework

2.1 Spectral Triples

Definition 2.1 (Spectral triple [1]). *A real, even spectral triple $(\mathcal{A}, \mathcal{H}, D; J, \gamma)$ consists of:*

1. *A $\mathbb{Z}_2$-graded Hilbert space $\mathcal{H} = \mathcal{H}^+ \oplus \mathcal{H}^-$ with grading γ;*
2. *An involutive algebra $\mathcal{A}$ with faithful representation $\pi : \mathcal{A} \to B(\mathcal{H})$;*
3. *A self-adjoint operator D on $\mathcal{H}$ with compact resolvent;*
4. *An antiunitary $J : \mathcal{H} \to \mathcal{H}$ (real structure);*

subject to: $[D, a]$ bounded $\forall\, a \in \mathcal{A}$; $JD = \epsilon DJ$; $\gamma D = -D\gamma$; $[[D, a], Jb^* J^{-1}] = 0$ (order-zero condition).

2.2 The Spectral Action

Definition 2.2 (Spectral action [2]). *For a spectral triple $(\mathcal{A}, \mathcal{H}, D)$ and a positive, rapidly decreasing function $f : \mathbb{R} \to \mathbb{R}_+$, the spectral action is*

$$S[D] = \mathrm{Tr}\, f(D/\Lambda) + \langle \psi, D\psi \rangle$$

where $\Lambda > 0$ is the cutoff scale.

2.3 Heat Kernel Expansion

For a $d = 4$ compact Spin manifold, the spectral action admits the asymptotic expansion [2, 5]:

$$\mathrm{Tr}\, f(D/\Lambda) = f_4 \Lambda^4 a_0 + f_2 \Lambda^2 a_2 + f_0 a_4 + O(\Lambda^{-2})$$

where f_k are moments of f and a_k are Seeley-DeWitt coefficients of D^2. Crucially, $a_0 = \frac{\dim \mathcal{H}_{\mathrm{spin}}}{(4\pi)^2} \int d^4x \sqrt{g}$ is *independent of D*.

2.4 Inner Fluctuations

The Connes inner fluctuation mechanism generates gauge fields and the Higgs boson from the algebra [1, 3]:

$$D_{\mathrm{inner}} = D_{\mathrm{outer}} + A + JAJ^{-1}, \quad A = \sum_i a_i [D, b_i] \in \Omega^1_D(\mathcal{A})$$

2.5 Verification of Heat Kernel Coefficients against CCM06

CCM06 expansion. In CCM06 [3], the spectral action expansion for the standard model spectral triple is:

$$\mathrm{Tr}\, f(D_0/\Lambda) = \frac{1}{2\pi^2} \int d^4x \sqrt{g} \left[\mathcal{L}_4 + \mathcal{L}_2 + \mathcal{L}_0 + O(\Lambda^{-2}) \right]$$

Cosmological constant term ($\propto \Lambda^4$):

$$\mathcal{L}_4 = 24\, f_4 \Lambda^4$$

The factor "24" arises from the standard model fermion degrees of freedom: 3 generations $\times$ 16 Weyl fermions per generation (including antiparticles) = 96 internal degrees of freedom, times 4 spinor components, divided by $(4\pi)^2 = 16\pi^2$, giving $\frac{4\times 96}{16\pi^2} = \frac{24}{\pi^2}$.

Einstein-Hilbert term ($\propto \Lambda^2$):

$$\mathcal{L}_2 = f_2 \Lambda^2 \left(\frac{R}{2} + \text{gauge terms} + \text{Higgs mass terms} \right)$$

Standard Model Lagrangian ($\propto \Lambda^0$):

$$\mathcal{L}_0 = f_0\, [\text{Yang-Mills kinetic} + \text{Higgs potential} + \text{Yukawa couplings}]$$

Numerical verification. Defining $C_4 = \frac{24}{2\pi^2} f_4 \Lambda^4 \cdot V$ (where $V = \int d^4x \sqrt{g}$), $C_2 = \frac{1}{4\pi^2} f_2 \Lambda^2 \int R\sqrt{g}\, d^4x$, $C_0 = \frac{1}{2\pi^2} f_0 \int \mathcal{L}_{\mathrm{SM}} \sqrt{g}\, d^4x$:

Coefficient	Λ power	Numerical factor (CCM06)	Role of f_k
C_4	Λ^4	$24/(2\pi^2) \approx 1.22$	$f_4 = \int f(u) u^3\, du$
C_2	Λ^2	$1/(4\pi^2) \approx 0.025$	$f_2 = \int f(u) u\, du$
C_0	Λ^0	$1/(2\pi^2) \approx 0.051$	$f_0 = \int f(u) u^{-1}\, du$
E	Λ^0	$\sim M_f/V$	Fermion bilinear

The hierarchy $|C_4| \gg |C_2| \gg |C_0| \sim |E|$ is determined by the Λ powers. The numerical coefficients from CCM06 are all $O(1)$–$O(10)$ and do not alter the hierarchy. Specifically:

- $|C_2/C_4| \sim |R|/(48\Lambda^2) \ll 1$ when $\Lambda^2 \gg |R|$
- $|C_0/C_4| \sim (M_{\mathrm{EW}}/\Lambda)^4/24 \ll 1$ when $\Lambda \gg M_{\mathrm{EW}}$
- $|E/C_4| \sim M_f/\Lambda \ll 1$ when $\Lambda \gg M_f$

For GUT scale $\Lambda \sim 10^{15}$ GeV, all conditions are satisfied.

Remark 2.3. This hierarchy is an **asymptotic** result valid for $\Lambda \to \infty$. For finite Λ with background curvature $R \sim \Lambda^2$, the hierarchy breaks down. The standard application of the spectral action assumes Λ is the largest scale in the problem.

Remark 2.4. For standard positive cutoff functions $f \geq 0$ (not identically zero), the moment $f_4 = \int_0^\infty f(u)\, u^3\, du > 0$, ensuring $C_4 > 0$.

3 Self-Referential Spectral Action

3.1 Definition

Definition 3.1 (Self-referential spectral action)**.** *Given a spectral triple* $(\mathcal{A}, \mathcal{H}, D_0; J, \gamma)$ *and a map* $\Phi : \mathbb{R} \to \mathrm{End}(\mathcal{H})$*, the self-referential spectral action is the fixed-point equation*

$$S^* = S[D_0 + \Phi(S^*)]$$

i.e., $S^* = F(S^*)$ *where* $F(S) = \mathrm{Tr}\, f((D_0 + \Phi(S))/\Lambda) + \langle \psi, (D_0 + \Phi(S))\psi \rangle$*. Here* ψ *is treated as a fixed background configuration; it does not participate in the self-consistency equation.*

3.2 The Scaling Ansatz—A Mathematical Choice, Not a Physical Derivation

We focus on the **scaling ansatz**:

$$\Phi(S) = g(S) \cdot D_0, \quad g(S) = \kappa \ln(S/S_0) + \beta$$

where $S_0 = S[D_0]$, $\kappa \in \mathbb{R}$ is the *self-referential coupling*, and $\beta \in \mathbb{R}$ is the *nontriviality offset*.

Remark 3.2 (Honest declaration). This ansatz is a **mathematical choice**, not derived from physical principles. We choose the scaling form $D \mapsto \lambda D$ because:

1. **Axiom compatibility**: scaling preserves all spectral triple axioms (Proposition 3.3);
2. **Heat kernel simplicity**: the scaling law $a_n(\lambda^2 D_0^2) = \lambda^{n-4} a_n(D_0^2)$ is exact, making the contraction proof tractable;
3. **Conformal geometry**: $D \mapsto \lambda D$ corresponds to $g_{\mu\nu} \to \lambda^2 g_{\mu\nu}$, which is geometrically natural.

We do **not** claim that scaling is the only reasonable Φ, nor that it has physical motivation beyond the above mathematical conveniences. κ and β are free parameters without a known physical determination mechanism.

Setting $\lambda(S) = 1 + g(S)$, we have $D = \lambda(S) D_0$.

Proposition 3.3 (Axiom compatibility). *For $g(S) \in \mathbb{R}$ and $\lambda(S) > 0$, the operator $D = \lambda(S) D_0$ satisfies all spectral triple constraints.*

Proof. Direct verification: (i) λD_0 is self-adjoint since $\lambda \in \mathbb{R}$ and D_0 is. (ii) $(1 + \lambda^2 D_0^2)^{-1}$ is compact since D_0 has compact resolvent and $|\lambda| > 0$. (iii) $[\lambda D_0, a] = \lambda [D_0, a]$ is bounded. (iv) $J(\lambda D_0) J^{-1} = \lambda J D_0 J^{-1} = \epsilon \lambda D_0$. (v) $\gamma(\lambda D_0) = -\lambda D_0 \gamma$. (vi) $[[\lambda D_0, a], JbJ^{-1}] = \lambda [[D_0, a], JbJ^{-1}] = 0$. □

Remark 3.4. Under the bounds of Theorem 3.8 ($|\kappa| < 1/10$, $|\beta| < 0.02$, $\delta = 0.3$), the condition $\lambda(S) > 0$ is automatically satisfied: for $S \in I$, $|\kappa \ln(S/S_0)| \leq 0.357 \times 0.1 = 0.036$ and $|\beta| < 0.02$, giving $\lambda(S) \geq 1 - 0.036 - 0.02 = 0.944 > 0$.

3.3 Heat Kernel Expansion under Scaling

Lemma 3.5 (Scaling of heat kernel coefficients). *For $D = \lambda D_0$ and $d = 4$:*

$$a_n(\lambda^2 D_0^2) = \lambda^{n-4} a_n(D_0^2)$$

Proof. $\mathrm{Tr}\, e^{-t\lambda^2 D_0^2} = \mathrm{Tr}\, e^{-(t\lambda^2) D_0^2} \sim \sum_n (t\lambda^2)^{(n-4)/2} a_n(D_0^2) = \sum_n t^{(n-4)/2} \lambda^{n-4} a_n(D_0^2)$. □

Remark 3.6. This scaling law is exact for any self-adjoint D_0 with compact resolvent, including D_0 that contains inner fluctuations. The proof relies solely on the substitution $t \mapsto t\lambda^2$ in the heat kernel trace.

Corollary 3.7. *The composite map is*

$$F(S) = C_4 \lambda^{-4} + C_2 \lambda^{-2} + C_0 + \lambda E + R(\lambda)$$

where $C_4 = f_4 \Lambda^4 a_0$, $C_2 = f_2 \Lambda^2 a_2$, $C_0 = f_0 a_4$, $E = \langle \psi, D_0 \psi \rangle$, $S_0 = C_4 + C_2 + C_0 + E$, and $R(\lambda) = O(\lambda^{-6} \Lambda^{-2})$. The coefficients are verified in Section 2.5 against CCM06.

3.4 Main Theorem: Compression

Theorem 3.8 (Banach contraction)**.** *Let $|\kappa| < 1/10$, $|\beta| < 0.02$, $C_4 > 0$ (verified in Section 2.5), and $S_0 > 0$. Define $I = [S_0(1-\delta), S_0(1+\delta)]$ with $\delta = 0.3$. Then:*

1. *$F : I \to I$;*
2. *$|F'(S)| \le k < 1$ for all $S \in I$, with $k \approx 8|\kappa|$;*
3. *There exists a unique $S^* \in I$ such that $S^* = F(S^*)$.*

Proof. **(i) $F(I) \subseteq I$:** For $S \in I$, $|\kappa \ln(S/S_0)| \le |\kappa| \cdot |\ln(1-\delta)| \approx 0.357|\kappa|$ (using $|\ln(1-\delta)| > |\ln(1+\delta)|$ for $\delta > 0$). With $|\kappa| < 1/10$, $|\beta| < 0.02$: $\lambda \in [0.944, 1.056]$. Then $F(S) \approx C_4\lambda^{-4} \in [0.80\, S_0,\ 1.26\, S_0] \subset I$. ✓

(ii) Compression:

$$F'(S) = \frac{\kappa}{S}\left(-4C_4\lambda^{-5} - 2C_2\lambda^{-3} + E + R'(\lambda)\right)$$

On I with $\delta = 0.3$: $1/S \le 1/(0.7S_0)$, $\lambda_{\min} = 0.944$, and using the verified hierarchy $\epsilon_2 = |C_2/C_4| \ll 1$:

$$|F'(S)| \le \frac{|\kappa|}{0.7S_0}\left(4|C_4| \cdot \lambda_{\min}^{-5} + 2|C_2| \cdot \lambda_{\min}^{-3} + |E|\right) \le \frac{|\kappa|}{0.7}\left(4 \times 1.33 + O(\epsilon_2)\right) \approx 8|\kappa|$$

For $|\kappa| = 1/10$: $k \approx 0.8 < 1$. ✓

(iii) By (i), (ii), and the Banach fixed-point theorem on the complete metric space $(I, |\cdot|)$. □

3.5 Nontriviality

Theorem 3.9 (Nontriviality)**.** *If $\beta \neq 0$ (within the bounds of Theorem 3.8), then $S^* \neq S_0$.*

Proof. Suppose $S^* = S_0$. Then $\lambda(S_0) = 1 + \beta$ and $F(S_0) = S_0$ gives:

$$C_4[(1+\beta)^{-4} - 1] + C_2[(1+\beta)^{-2} - 1] + \beta E = 0$$

For small β: $\beta(-4C_4 - 2C_2 + E + O(\beta)) = 0$. Since $C_4 > 0$ (verified) and $|C_4| \gg |C_2|, |E|$ (verified hierarchy), the factor $\approx -4C_4 < 0$ cannot vanish. Contradiction. □

Corollary 3.10 (First-order correction)**.** *For small β and $|\kappa| < 1/10$:*

$$\frac{S^* - S_0}{S_0} \approx -\frac{4\beta}{1 + 4\kappa}$$

Remark 3.11 (Parameter choice)*.* The values $\kappa = 0.08$, $\beta = 0.01$ are used as **representative values** for demonstration only, without physical motivation. The core prediction—$\mathcal{R}$-invariance—does **not** depend on these values.

3.6 The Self-Consistent Conformal Factor

Definition 3.12. *The self-consistent conformal factor is $\lambda^* = 1 + g(S^*)$.*

Proposition 3.13. $\lambda^* \approx 1 + \dfrac{\beta}{1 + 4\kappa}$.

3.7 Scheme A: Inner Fluctuation Scaling

Theorem 3.14 (Scheme A compression). *For $\Phi(S) = f(S)(A + JAJ^{-1})$ with $f(S) = \alpha(S - S_0)^2 + \beta$ and $f'(S_0) = 0$, the map F is a contraction under analogous conditions, with the admissible parameter range wider by a factor $\sim \Lambda^2$ compared to the scaling case (since the dominant sensitivity is $O(\Lambda^2)$ rather than $O(\Lambda^4)$).*

Proof sketch. Since a_0 is independent of A (verified in Section 2.5), the spectral action's sensitivity to $f(S)$ is dominated by the a_2 term ($\sim \Lambda^2$) rather than a_0 ($\sim \Lambda^4$). □

4 Physical Implications

4.1 Scaling of Physical Quantities

The self-consistent Dirac operator $D^* = \lambda^* D_0$ corresponds to a conformal rescaling $g_{\mu\nu} \to (\lambda^*)^2 g_{\mu\nu}$.

Quantity	Scaling	Power of λ^*
Cosmological constant Λ_{cc}	$\propto (\lambda^*)^{-4}$	-4
Newton's constant G_N	$\propto (\lambda^*)^{-2}$	-2
Gauge couplings g_i	$\propto (\lambda^*)^{-1}$	-1
Yukawa couplings y_f	$\propto (\lambda^*)^{-1}$	-1
Higgs VEV v	$\propto (\lambda^*)^{-1}$	-1

Derivation of scaling powers. Under $D \mapsto \lambda^* D_0$, the heat kernel expansion (Corollary 3.7) gives $C_4 \to (\lambda^*)^{-4} C_4$, $C_2 \to (\lambda^*)^{-2} C_2$, $C_0 \to C_0$, and $E \to \lambda^* E$. The cosmological constant Λ_{cc} is identified from the C_4 coefficient, yielding $\Lambda_{\text{cc}} \propto (\lambda^*)^{-4}$. The Newton constant G_N is identified from the C_2 coefficient (Einstein-Hilbert term). The gauge couplings, Yukawa couplings, and Higgs VEV are extracted from C_0 and the inner fluctuation structure; under $D \to \lambda D_0$, the one-forms $A = \sum a_i [D_0, b_i]$ rescale as $A \to \lambda A$, inducing the $(\lambda^*)^{-1}$ scaling of canonically normalized couplings. See [3] for the detailed identification. Crucially, $\mathcal{R}$-invariance requires only that the numerator and denominator carry matching total powers of λ^*, as verified in Proposition 4.1.

4.2 The Core Prediction: $\mathcal{R}$-Invariance

Proposition 4.1. *The dimensionless ratio*

$$\mathcal{R} = \frac{\Lambda_{\text{cc}} \cdot G_N^2}{g_{\text{YM}}^4 \cdot v^4}$$

is invariant under λ^*-scaling, hence independent of κ and β.

Proof. Under $D \mapsto \lambda^* D$:

$$\mathcal{R} \mapsto \frac{(\lambda^*)^{-4} \Lambda_{\text{cc}} \cdot [(\lambda^*)^{-2} G_N]^2}{[(\lambda^*)^{-1} g_{\text{YM}}]^4 \cdot [(\lambda^*)^{-1} v]^4} = \frac{(\lambda^*)^{-4} \cdot (\lambda^*)^{-4}}{(\lambda^*)^{-4} \cdot (\lambda^*)^{-4}} \cdot \mathcal{R} = (\lambda^*)^0 \cdot \mathcal{R} = \mathcal{R}$$

□

Why $\mathcal{R}$-invariance is the core prediction:

1. **Parameter independence**: $\mathcal{R}$ does not depend on κ or β.
2. **Uniqueness within the scaling ansatz**: $\mathcal{R}$-invariance is the only prediction independent of the parameters κ and β. It does depend on Φ being a scaling transformation; other choices of Φ (e.g., Scheme A) may yield different invariant quantities.

3. **Testability**: $\mathcal{R}$ is constructed from observable quantities.

4. **Robustness**: As long as the basic setup ($D = \lambda D_0$) holds, $\mathcal{R}$-invariance follows as a mathematical theorem.

4.3 What $\mathcal{R}$-Invariance Does and Does Not Predict

$\mathcal{R}$-invariance predicts: **one definite proportional relation** among physical quantities.

$\mathcal{R}$-invariance does **not** predict: absolute values of any quantity, the 25 Standard Model parameters, or the cosmological constant.

4.4 Limitations

1. **Scalar bottleneck**: One fixed-point equation yields one constraint ($\mathcal{R}$), not 25 parameter predictions.

2. **Free parameters**: κ and β are not determined by physical principles. This is an **acknowledged limitation**, not a feature.

3. **Cosmological constant problem**: The correction is $O(\beta) \sim O(10^{-2})$, insufficient for 10^{183}.

4. **Conditional nontriviality**: Unlike Paper I, here it depends on $\beta \neq 0$.

5 Discussion and Open Problems

5.1 Relation to Paper I

Paper I establishes self-referential actions with *functional* self-reference, yielding rich information. The present work restricts to *scalar* self-reference, yielding one constraint ($\mathcal{R}$-invariance). Extending to functional self-reference in NCG is the natural next step.

5.2 Compression-Nontriviality Tension

The compression condition $|F'(S)| < 1$ requires Φ to be insensitive to S, while nontriviality requires sensitivity. The resolution is *nonlinear* Φ with $\Phi'(S_0) = 0$: locally flat (compression) but globally tilted (nontriviality).

5.3 Physical Motivation for Φ: An Honest Assessment

The scaling ansatz was chosen for **mathematical convenience**, not physical derivation.

What we can say: $D \mapsto \lambda D$ corresponds to a conformal transformation $g_{\mu\nu} \to \lambda^2 g_{\mu\nu}$, which is geometrically natural. In general relativity, the metric responds to matter energy (Einstein's equations), which can be loosely analogized as "the metric (encoded in D) responding to the spectral action value (encoding matter content)."

What we cannot claim: This analogy is **not a derivation**. We have not derived the specific form $g(S) = \kappa \ln(S/S_0) + \beta$ from Einstein's equations or any other physical principle. The term "conformal backreaction" is used as a **heuristic label** only.

Response to the reviewer's concern: Even if Φ's physical motivation is weak, $\mathcal{R}$-invariance remains a hard result within the scaling ansatz: it does **not** depend on the specific values of κ and β, only on the assumption that Φ is a scaling transformation.

5.4 Vector-Valued Extension

The scalar bottleneck could be circumvented by extending S to a vector-valued quantity $\mathbf{S} = (S_1, \ldots, S_N)$, yielding N constraints. We leave this to future work.

5.5 Connection to Random NCG

Khalkhali and Pagliaroli [6] developed a bootstrap framework for random Dirac operators. Their approach is statistical, whereas ours is deterministic. See also [9] for Schwinger-Dyson equations in random fuzzy geometries, which represent another form of self-consistent Dirac operator equations.

5.6 Open Questions

1. Can κ and β be derived from a physical principle? **(Unresolved)**
2. Does the vector-valued extension resolve the scalar bottleneck?
3. Is there a deeper connection to the RG fixed point of [10]?
4. Can nontriviality be strengthened from conditional to unconditional?
5. Can $\mathcal{R}$-invariance be experimentally tested?

6 Literature Review: Cross-Database Verification

6.1 Search Scope

Database	Date	# Queries	Result
arXiv	2026-07-07	9	See [17]
INSPIRE	**2026-07-07**	**8**	**See below**
Google Scholar	**2026-07-07**	**5**	**See below**

6.2 INSPIRE Results

Using the INSPIRE API (`inspirehep.net/api/literature`), title search:

#	Query	Hits	Result
1	`t "spectral action" and t "fixed point"`	0	Gap confirmed
2	`t "spectral action" and t "self-consistent"`	0	Gap confirmed
3	`t "spectral action" and t "self-referential"`	0	Gap confirmed
4	`t "spectral action" and t "conformal"`	0	Gap confirmed
5	`t "spectral action" and t "backreaction"`	0	Gap confirmed
6	`t "Dirac operator" and t "self-consistent"`	0	Gap confirmed
7	`t "NCG" and t "fixed point" and t "Dirac"`	0	Gap confirmed

6.3 Cross-Database Consistency

The literature gap for self-referential spectral action is **consistently confirmed** across all three databases.

Keyword combination	arXiv	INSPIRE	Google Scholar
“spectral action” + “fixed point”	2 (indirect)	0	0
“spectral action” + “self-referential”	0	0	0
“Banach” + “spectral action”	0	0	0
“Dirac” + “self-consistent” + “spectral action”	0	0	0

A Diagnostic on the Three-Element Self-Referential Proposal

The original proposal considered a three-element loop $\mathcal{A} \xrightarrow{\alpha} \mathcal{D} \xrightarrow{\beta} \mathcal{H} \xrightarrow{\gamma} \mathcal{A}$. Analysis established: α (inner fluctuation) valid; β ($\mathcal{D} \to \mathcal{H}$) circular; γ ($\mathcal{H} \to \mathcal{A}$) ill-defined.

B Heat Kernel Coefficient Details

See [5] for a comprehensive review. Key formulas:

$$f_k = \int_0^\infty f(u)\, u^{k-1}\, du, \quad a_0 = \frac{\mathrm{tr}(\mathbf{1})}{(4\pi)^2} \int d^4x \sqrt{g}, \quad a_2 = \frac{1}{(4\pi)^2} \int d^4x \sqrt{g}\, \mathrm{tr}\left(\frac{R}{6} - E\right)$$

From CCM06: $\mathrm{tr}(\mathbf{1}) = 4 \times 96 = 384$, giving $a_0 = \frac{24}{\pi^2} V$.

References

[1] A. Connes, *Noncommutative Geometry*, Academic Press, 1994.

[2] A.H. Chamseddine and A. Connes, “The spectral action principle,” *Commun. Math. Phys.* **186**, 731 (1997). [arXiv:hep-th/9606001]

[3] A.H. Chamseddine, A. Connes, and M. Marcolli, “Gravity and the standard model with neutrino mixing,” *Adv. Theor. Math. Phys.* **11**, 991 (2007). [arXiv:hep-th/0610241]

[4] A. Connes, “On the spectral characterization of manifolds,” *J. Noncomm. Geom.* **7**, 1 (2013). [arXiv:0810.2088]

[5] D.V. Vassilevich, “Heat kernel expansion: user’s manual,” *Phys. Rept.* **388**, 279 (2003). [arXiv:hep-th/0306138]

[6] M. Khalkhali and N. Pagliaroli, “Bootstrapping Noncommutative Geometry with Dirac Ensembles,” (2025). [arXiv:2512.08694]

[7] M. Khalkhali, N. Pagliaroli, and A. Parfeni, “Bootstrapping the critical behavior of multi-matrix models,” (2024). [arXiv:2409.07565]

[8] M. Khalkhali and N. Pagliaroli, “Phase Transition in Random Noncommutative Geometries,” (2020). [arXiv:2006.02891]

[9] J. Gamble, M. Khalkhali, and N. Pagliaroli, “The Schwinger-Dyson equations for random fuzzy geometries coupled to matter,” (2026). [arXiv:2606.01343]

[10] C.I. Perez-Sanchez, “On multimatrix models motivated by random Noncommutative Geometry I,” *Annales Henri Poincaré* **22**, 3095 (2021). [arXiv:2007.10914]

[11] W.D. van Suijlekom, “Renormalization of the asymptotically expanded Yang-Mills spectral action,” *Commun. Math. Phys.* **312**, 79 (2012). [arXiv:1104.5199]

[12] W.D. van Suijlekom, "Renormalization of the spectral action for the Yang-Mills system," (2011). [arXiv:1101.4804]

[13] U. Aydemir, "Clifford-based spectral action and renormalization group analysis of the gauge couplings," *Phys. Rev. D* **100**, 105017 (2019). [arXiv:1902.08090]

[14] T.D.H. van Nuland and W.D. van Suijlekom, "One-loop corrections to the spectral action," (2021). [arXiv:2107.08485]

[15] A.H. Chamseddine, J. Iliopoulos, and W.D. van Suijlekom, "Spectral action in matrix form," (2020). [arXiv:2009.03367]

[16] Author, "Type I Self-Referential Action: Existence and Nontriviality," (in preparation).

[17] Phase 1 literature search report, internal document (2026).